Proceedings of the ASME 2026
International Mechanical Engineering Congress and Exposition
IMECE2026
November 8-12, 2026, Vancouver, British Columbia, Canada

IMECE2026

TRANSIENT THERMAL ANALYSIS OF ACTIVE AND PASSIVE INTERVENTIONS FOR NEXT-GENERATION WILDLAND FIREFIGHTER EMERGENCY SHELTERS

Ana Gegetchkori[1], Francisca Gomez Weiss[1], Kristian Nasi[1], Jaylen Patel[1], Chris Schaefer[1], Ayden Wells[1]

[1]Department of Aerospace and Mechanical Engineering, Saint Louis University
St. Louis, MO, USA

ABSTRACT

This study evaluates passive Phase Change Material (PCM) and active CO_2 heat exchanger interventions to enhance wildland firefighter emergency shelter survivability under 2192°F (1200°C) burn-over conditions. Using a transient lumped-parameter model, the intervention designs were evaluated against the standard M-2002 shelter, which takes 11.8 s to reach the 122°F (49°C) interior temperature threshold associated with human thermal tolerance. A Paraffin RT42 PCM layer delayed this limit to 63.1 s, adding 4.66 kg of mass, while a single-pass CO_2 manifold delayed it to 39.1 s with a 1.10 kW peak heat removal. Results establish key design trade-offs between latent heat capacity, cooling limits, and firefighter pack weight with added mass.

1. INTRODUCTION

This paper presents a preliminary analysis of two emergency wildland fire shelter interventions designed to enhance thermal buffering and environmental safety without exceeding standard wildland firefighter pack limits for mass, stowed volume, manufacturing and implementation costs, and deployment ergonomics. Global wildfires are increasing in frequency, intensity, and burned acreage, with approximately 112.3 million hectares burned globally in 2026 [1]. These catastrophic events coincide with a 20-year U.S. high in firefighter entrapment incidents [2]. Wildland firefighters face increasingly severe operational hazards, including direct flame impingement, thermal entrapment during suppression operations, and radiant heat fluxes exceeding 277 kW/m^2 from fire temperatures reaching 2192°F (1200°C) [3, 4].

During critical burn-over events, standard emergency fire shelters serve as the last line of defense, providing protection by reflecting initial radiant heat with an aluminum foil-laminate outer layer; however, minimal thermal buffering is provided once heat penetrates the outer barrier. Over a typical 30-minute entrapment exposure, thermal energy, metabolic heat, and exhaled carbon dioxide (CO_2), can accumulate rapidly within the small, sealed interior volume. The life-threatening consequence of this thermal limitation was demonstrated in the 2013 Yarnell Hill Fire in Arizona, where 19 firefighters lost their lives despite deploying their emergency shelters [5]. Existing efforts to improve shelter performance have focused on decreasing weight and adding insulation materials, leaving latent heat buffering and active gas-based heat removal as potential areas for further investigation [6].

2. MATERIALS AND METHODS

A comparison of two distinct intervention designs was conducted against the baseline standard fire shelter, the M-2002 Fire Shelters [7], with each targeting a different mechanism of heat management. The PCM Fire Shelter integrates a passive Phase Change Material (PCM), specifically a Paraffin RT42 layer selected for its latent heat of fusion of 165 kJ/kg and melting temperature range of 35°C – 50°C [8]. Heat penetrates the outer foil and is absorbed as latent heat during the phase change transition from solid to liquid, maintaining the PCM near its melting range until the phase change is complete. The PCM wax layer acts as a thermal battery, absorbing heat and extending the duration the interior air stays below 122°F (49°C). The CO_2 Heat Exchanger Shelter attaches a CO_2 tank to a manifold that routes the gas into four quarter-inch tubes. The tubes run in a single-pass configuration along the top of the fire shelter with fins attached, acting as a heat exchanger before venting outside the shelter. The CO_2 flow absorbs heat from the shelter through the tube walls and attached fins, with the cooling capacity determined by the CO_2 mass flow rate and thermodynamic properties.

Each concept is evaluated using a transient lumped-parameter model with a 6-node baseline and 7-node PCM and CO_2 models, with the air gaps modeled as thermal resistances. The shelter base is modeled as a constant-temperature heat sink representing contact with the ground. For the PCM concept, the nodes include the outer radiant layer, the PCM layer, and the internal liner. For the CO_2 heat exchanger, the CO_2 transfer layer replaces the PCM layer. Heat transfer from the external environment was modeled using Stefan-Boltzmann radiation and forced convection, while conduction between layers is calculated

© 2026 by ASME

through the corresponding thermal resistances. The predicted transient temperatures were used to compare the time required for each design to reach an interior temperature of 122°F (49°C) under a constant incident heat flux of 277 kW/m^2. Each transient thermal model was implemented in MATLAB using the ode45 ordinary differential equation solver [10].

3. RESULTS AND DISCUSSION

Both concepts were evaluated under a constant external fire temperature of 2192°F (1200°C) for 30 minutes, representing severe thermal exposure conditions. Under these conditions, the interior air of the baseline M-2002 shelter reached 122°F (49°C) after 11.8 s. The Paraffin RT42 PCM layer extended the time to reach the 122°F (49°C) interior limit to 63.1 s. The PCM melted completely at 33.7 s, after which the interior temperature rose rapidly, indicating that the latent heat capacity of the layer limits the improvement, rather than its conductive resistance. The 4.66 kg increase in shelter mass establishes a trade-off between protection duration and shelter mass. Furthermore, the CO_2 Heat Exchanger Shelter delayed the interior air temperature from reaching 122°F (49°C) to 39.1 s, representing an improvement over the baseline shelter, but a shorter protection duration than the PCM configuration. The CO_2 system provided a maximum modeled heat removal rate of 1.10 kW for approximately 331 s of continuous operation. Despite the active cooling, the interior air reached 1003°F (539°C) after 30 minutes. These preliminary results demonstrate that both concepts increased the time before the interior reached the 122°F (49°C) design limit compared with the baseline shelter; however, each concept has a different limitation. The PCM concept is constrained by latent-heat capacity and mass, while the CO_2 system is constrained by cooling capacity and available CO_2 mass.

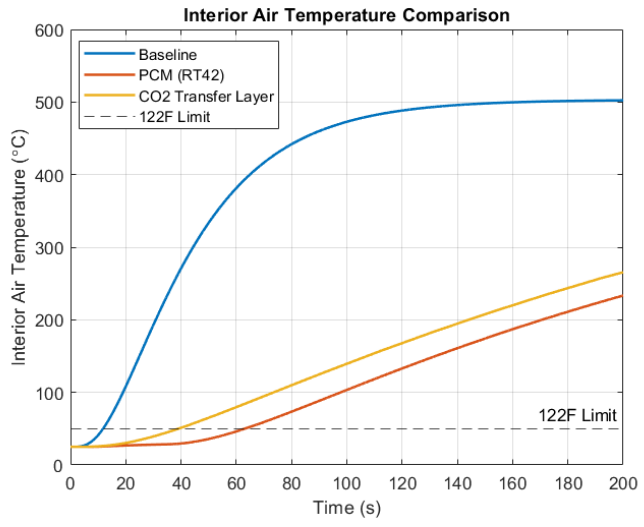


FIGURE 1: INTERIOR AIR TEMPERATURE RESPONSE MODEL COMPARISON

4. CONCLUSION

This preliminary analysis demonstrates that both the PCM and CO_2 heat exchanger concepts extend the time before the shelter interior reaches the 122°F (49°C) thermal limit compared to the baseline M-2002 shelter. The PCM configuration provides the longest protection duration but requires additional mass and is limited by its latent-heat capacity. The CO_2 system provided active cooling but was limited by its available cooling capacity and supply. Future work will refine the thermal models, assess the designs under varying exposure conditions, and investigate different geometric configurations and layer arrangements.

REFERENCES

[1] Visual Capitalist, 2026, “Where the World Is Burning in 2026,” Visual Capitalist, www.visualcapitalist.com/where-the-world-is-burning-in-2026/ [Accessed: Sep. 20, 2026].

[2] Reuters, 2026, “Record U.S. Wildfire Season, Firefighter Entrapments Hit 20-Year High,” Reuters, www.reuters.com/business/environment/record-us-wildfire-season-firefighter-entrapments-hit-20-year-high-2026-08-13/ [Accessed: Sep. 20, 2026].

[3] Carballo-Leyenda, B., Villa, J. G., López-Satué, J., and Rodríguez-Marroyo, J. A., 2019, “Characterizing Wildland Firefighters’ Thermal Environment During Live-Fire Suppression,” Front. Physiol., 10, p. 949, www.frontiersin.org/journals/physiology/articles/10.3389/fphys.2019.00949/full [Accessed: Sep. 20, 2026].

[4] U.S. Geological Survey, 2024, “How Hot Are Wildfires?” U.S. Geological Survey, www.usgs.gov/media/images/how-hot-are-wildfires [Accessed: Sep. 20, 2026].

[5] McCrea, R. C., 2014, "The Yarnell Hill Fire: A Review of Lessons Learned," Wildfire Magazine, 23(3), www.iawfonline.org/article/the-yarnell-hill-fire-a-review-of-lessons-learned/ [Accessed: September 28, 2026].

[6] Textile Technology Source, 2022, “New Fire Shelters Could Better Protect Firefighters,” Textile Technology Source, textiletechsource.com/2022/07/11/new-fire-shelters-could-better-protect-firefighters/ [Accessed: Sep. 20, 2026].

[7] National Wildfire Coordinating Group, 2026, “NWCG Standards for M-2002 Fire Shelters,” Publication PMS 411, National Wildfire Coordinating Group, www.nwcg.gov/publications/pms411/nwcg-standards-for-m-2002-fire-shelters-pms-411 [Accessed: Sep. 28, 2026].

[8] Rubitherm Technologies GmbH, “RUBITHERM® RT42: Data Sheet,” version 21.01.2026, Berlin, Germany, 2026.

[9] GoodRx, 2024, “What Temperature Is Too Hot for Humans? Understanding Heat Index,” GoodRx, www.goodrx.com/health-topic/environmental/temperature-too-hot-for-humans [Accessed: Sep. 20, 2026].

[10] MathWorks, Inc., *MATLAB*, Version R2025b, Natick, MA, 2025.

© 2026 by ASME